Decoherence Free Subsystems/Subspaces

- control of decoherence via Subspace/Subsystem selection
- collective decoherence eliminable - condensed phase QC/QM
- fault tolerant universal quantum computation with minimal decoherence
- new paradigm for universal QC via (DFS) encodings
- E.g., universal QC with exchange interaction only - condensed phase
- optimal gate constructions - e.g. 19-gate exchange-only CNOT

Future Goals:

- develop universal computation from system specific interactions
- optimization/minimization of gate interactions
- simulation of computation - quantum trajectories
- develop optical implementations of DFS - QC in the dark

DF Subspaces and Subsystems:

Zanardi 1997, Duan and Guo 1997, Lidar et al. 1998, E. Knill et al. 2000



Interaction causing DECOHERENCE

System evolution Bath evolution

$$H = H_S \otimes I_B + I_S \otimes H_B + \sum_\alpha S_\alpha \otimes B_\alpha$$

Expansion into system and bath operators

The S_α form an algebra (add, multiply the elements) which is reducible:

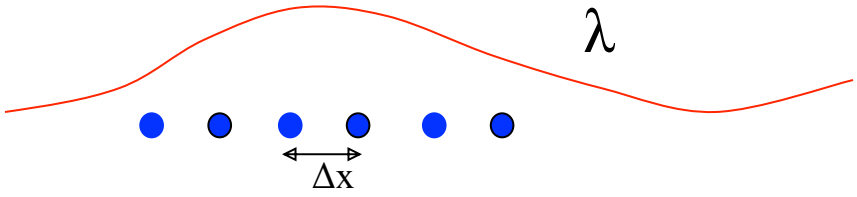
$$\mathcal{A} = \bigoplus_{J=1}^{r} \left(I_{n_J} \otimes M_{d_J} \right)$$

linear operators on a d_J dimensional space

I_d is the $d \times d$ identity operator n_J is the degeneracy of the Jth irrep

Hilbert space over the degenerate identity component is immune to the effects of decoherence! The quantum information then inhabits a decoherence-free subsystem (subspace if $d_J = 1$).

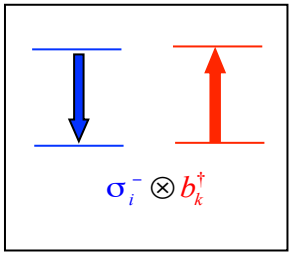
Collective Decoherence:



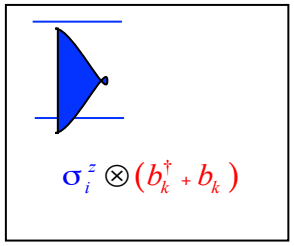
bath boson mode, λ

System Bath Interaction (spin-boson model): - Dissipative

$$H_{S-B} = \sum_{i=1}^{N} \sum_{\mathbf{k}} \underbrace{g^{+}_{i,\mathbf{k}} \sigma^{+}_{i} b_{\mathbf{k}} + g^{-}_{i,\mathbf{k}} \sigma^{-}_{i} b^{\dagger}_{\mathbf{k}}}_{\text{dissipation}} + \underbrace{g^{z}_{i,\mathbf{k}} \sigma^{z}_{i} \left(b_{\mathbf{k}} + b^{\dagger}_{\mathbf{k}} \right)}_{\text{dephasing}}$$



- Dephasing



Collective Interaction Approximation:

bath does not distinguish system particles

$$\boxed{g_{i,\mathbf{k}} \equiv g_{\mathbf{k}}} \longleftarrow \lambda \ldots \Delta x$$

particle permutation invariance

Define **collective** "angular momentum" operators:

$$S^{\alpha} = \sum_{i=1}^{N} \sigma_i^{\alpha} \qquad \alpha = +, -, z$$

satisfy **global** angular momentum *su(2)* algebra

$$H_{S-B} = S^{+} \otimes \left(\sum_{\mathbf{k}} g^{+}_{\mathbf{k}} b_{\mathbf{k}} \right) + S^{-} \otimes \left(\sum_{\mathbf{k}} g^{-}_{\mathbf{k}} b^{\dagger}_{\mathbf{k}} \right) + S^{z} \otimes \left(\sum_{\mathbf{k}} g^{z}_{\mathbf{k}} \left(b_{\mathbf{k}} + b^{\dagger}_{\mathbf{k}} \right) \right)$$

examples/candidates:

quantum optics "**Dicke subradiance**"

solid state systems at low temperatures,

atoms trapped in a cavity, etc.

Collective DFS Examples:

3 qubit strong collective decoherence, decoherence-free subsystem:

S^x over the degenerate portion $=$

$$\begin{pmatrix} 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 \end{pmatrix} \rightarrow I_2 \otimes s_x^2$$

$$\left|J=\tfrac{1}{2}, m_J=\tfrac{1}{2}\right\rangle = \tfrac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right)|0\rangle$$

$$\left|J=\tfrac{1}{2}, m_J=-\tfrac{1}{2}\right\rangle = \tfrac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right)|1\rangle$$

$$\left|J=\tfrac{1}{2}, m_J=\tfrac{1}{2}\right\rangle = \sqrt{\tfrac{2}{3}}|00\rangle|1\rangle - \sqrt{\tfrac{1}{3}}\left\{\tfrac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right)\right\}|0\rangle$$

$$\left|J=\tfrac{1}{2}, m_J=-\tfrac{1}{2}\right\rangle = \sqrt{\tfrac{2}{3}}|11\rangle|0\rangle - \sqrt{\tfrac{1}{3}}\left\{\tfrac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right)\right\}|1\rangle$$

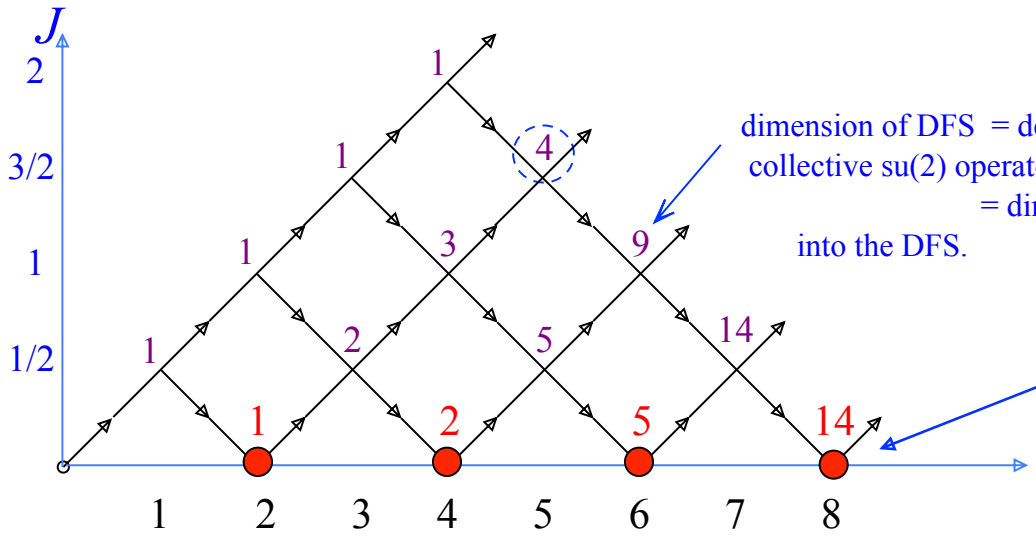
Information encoded over the degenerate component will not decohere!

4 qubit collective decoherence, decoherence-free subspace:

$$\left|J=0, m_J=0, n=0\right\rangle = \tfrac{1}{2}\left(|01\rangle - |10\rangle\right)\otimes\left(|01\rangle - |10\rangle\right)$$

$$\left|J=0, m_J=0, n=1\right\rangle = \tfrac{1}{\sqrt{3}}\left[|0011\rangle + |1100\rangle - \left(|01\rangle + |10\rangle\right)\otimes\left(|01\rangle + |10\rangle\right)\right]$$

Information encoded over the subspace will not decohere



dimension of DFS = degeneracy of the n qubit, total angular momentum J, collective su(2) operators

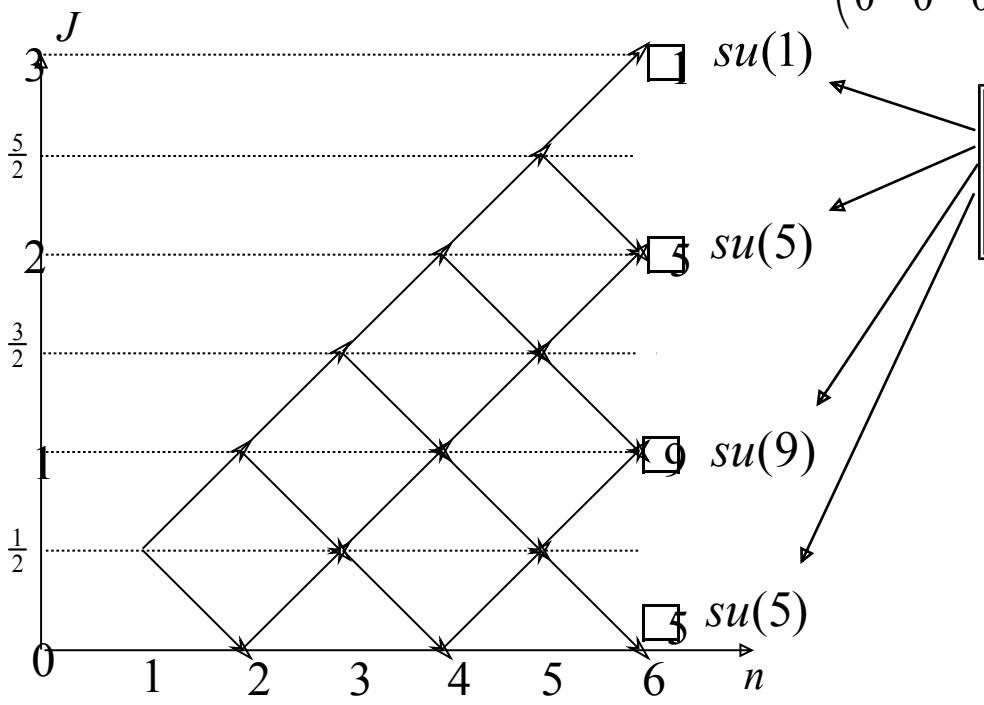
= dimension of quantum information which can be encoded into the DFS.

The J=0 irreps of su(2) are 1- dimensional

Decoherence Free Subspaces.

Quantum Computation on a DFS:

The Exchange Hamiltonian between 2 qubits $\equiv E = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} |00\rangle \\ |01\rangle \\ |10\rangle \\ |11\rangle \end{matrix}$. $E_{ij} \equiv$ E acting between qubits i and j.



Exchange can be used to perform su(d)'s on each of the DFSs.

Each su(n) can be enacted when the others turned off (**independent** action on each J irrep)

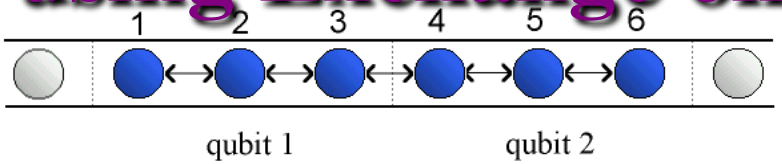
su(d) means we have a d-dimensional Hilbert space (encoded) and we can perform all unitary manipulations on this d-dimensional Hilbert space

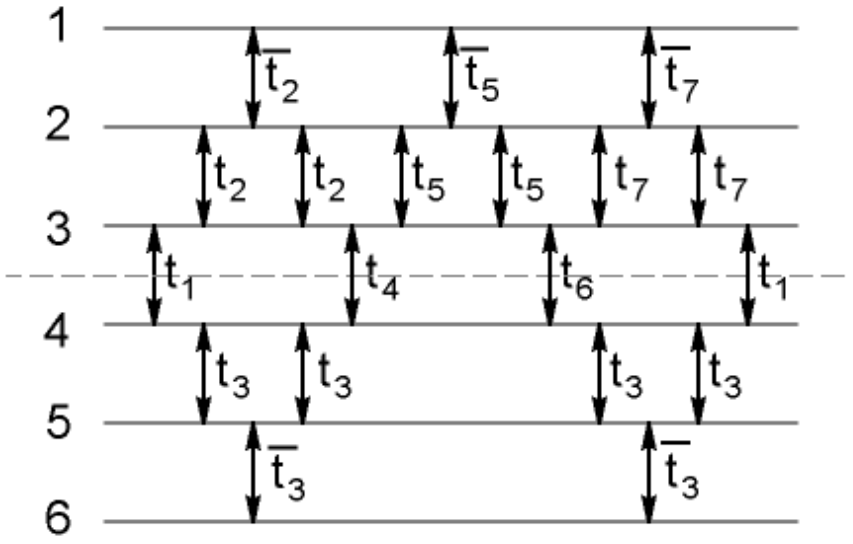
Kempe et. al. quant-ph/0004064

Exchange alone produces fault tolerant quantum computation on collective decoherence-free subsystems!!

system never leaves the code space during entire gate switching time

Explicit CNOT Gate Construction using Exchange only





t_1=0.410899(2) t_5=0.414720(10)

t_2=0.207110(20) t_6=0.147654(12)

t_3=0.2775258(12) t_7=0.813126(12)

t_4=0.640505(8) $\tan(\pi t)\tan(\pi \bar{t}) = -2$

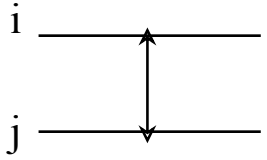
Heisenberg exchange coupling

$$H_{heis} = J(t)\, s^{(i)} \cdot s^{(j)}$$

$s^{(i)} = (s_x^{(i)}, s_y^{(i)}, s_z^{(i)})$ single spin operators on qubit i

central to quantum dot, Si, Si/Ge proposals

exchange gate



$= \exp(iJE_{i,j}t)$

DiVincenzo et al., quant-ph/0005116